Cambridge (CIE) A Level Maths: Probability & Statistics 1

Your notes

Working with Distributions

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Modelling with Distributions

When should I use a binomial distribution?

- A random variable that follows a **binomial distribution** is a **discrete random variable**
- A binomial distribution is used when the random variable **counts something**
  - The number of successful trials
  - The number of members of a sample that satisfy a criterion (satisfying the criteria can be seen as a successful trial)
- There are **four conditions** that X must fulfil to follow a binomial distribution
  - There is a **fixed finite number of trials (n)**
  - The trials are **independent**
  - There are **exactly two outcomes** of each trial (**success or failure**)
  - The **probability of success (p) is constant**

When should I use a geometric distribution?

- A random variable that follows a **geometric distribution** is a **discrete random variable**
- A geometric distribution is used when the random variable **counts something**
  - The number of trials until a successful trial
- The conditions that X must fulfil to follow a geometric distribution are exactly the same as for a binomial distribution except there is **no fixed number of trials**
  - Instead, the trials will continue until the **first time a success occurs**

When should I use a normal distribution?

- A random variable that follows a **normal distribution** is a **continuous random variable**
- A normal distribution is used when the random variable **measures something** and the distribution is:
  - **Symmetrical**
  - **Bell-shaped**
- A normal distribution can be used to model real-life data provided the **histogram** for this data is **roughly symmetrical and bell-shaped**
  - If the variable is normally distributed then as more data is collected the outline of the histogram should get smoother and resemble a normal distribution curve

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Can the binomial distribution and the normal distribution be used in the same question?

- Some questions might require you to **first use the normal distribution** to find the **probability of success** and then use the **binomial distribution**
  - Remember a discrete distribution is either a binomial or geometric distribution
- The key is to make sure you are very clear about what each **parameter/variable represents**

Worked Example

In a population of cows, the masses of the cows can be modelled using a normal distribution with mean 550 kg and standard deviation 80 kg. A farmer classifies cows as *beefy* if they weigh more than 700 kg. The farmer takes a random sample of 10 cows and weighs them.

Find the probability that at most one cow is *beefy*.

State the variables and distributions

Let M be the mass of a cow

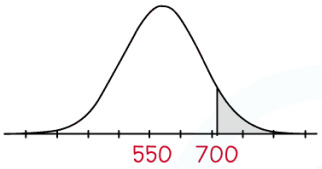
$M \sim N(550, 80^2)$

Let X be the number of beefy cows in the sample

$X \sim B(10, p)$ where p is the probability that a cow is beefy

Calculate the value of p

$p = P(M > 700)$



Standardise: $P(M > 700) = P\left(Z > \frac{700 - 550}{80}\right)$

$= P(Z > 1.875)$

$= 1 - \Phi(1.875)$

$= 1 - 0.9697$

Use table of values: $p = 0.0303$

Find the probabilty of at most 1 beefy cow

$X \sim B(10, 0.0303)$

$P(X \leqslant 1) = P_0 + P_1$

$\binom{10}{1} = 10$

$\binom{10}{0} = 1$

Using formula

$Pr = \binom{n}{r} P^r (1 - p)^{n-r}$

$= (0.9697)^{10} + 10(0.0303)(0.9697)^9$

$= 0.96485...$

0.965 (3sf)

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Examiner Tips and Tricks

- Always state what your variables and parameters represent. Make sure you know the conditions for when each distribution is (or is not) a suitable model.

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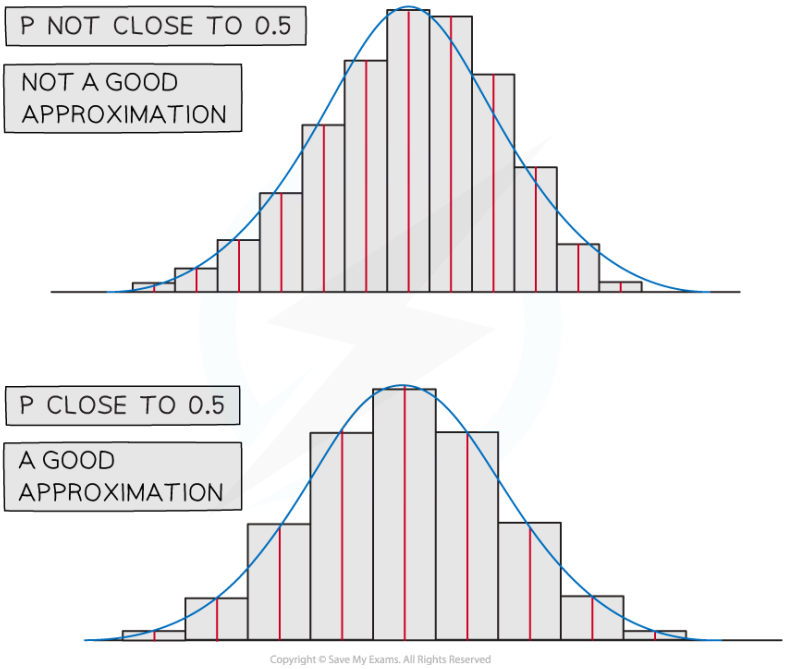
Normal Approximation of Binomial

When can I use a normal distribution to approximate a binomial distribution?

- A binomial distribution $X \sim \mathrm{B}(n,p)$ can be **approximated** by a normal distribution $X_N \sim \mathrm{N}(\mu,\sigma^2)$ provided
  - ***n* is large**
  - ***p* is close to 0.5**
    - $np > 5$
    - $nq > 5 \text{ where } q = 1 - p$
- The mean and variance of a binomial distribution can be calculated by:
  - $\mu = np$
  - $\sigma^2 = np(1-p)$



Why do we use approximations?

- If there are a large number of values for a binomial distribution there could be a lot of calculations involved and it is inefficient to work with the binomial distribution

- These days calculators can calculate binomial probabilities so approximations are no longer necessary
- However it is **easier** to work with a normal distribution
  - You can calculate the probability of a range of values quickly
  - You can use the inverse normal distribution function (most calculators don't have an inverse binomial distribution function)
- In your exam you must use the formula and not a calculator to find binomial probabilities so you are limited to small values of *n*

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What are continuity corrections?

- The **binomial distribution is discrete** and the **normal distribution is continuous**
- A **continuity correction** takes this into account when using a normal approximation
- The probability being found will need to be changed from a discrete variable, X, to a continuous variable, X_N
  - For example, $X = 4$ for binomial can be thought of as $3.5 \le X_N < 4.5$ for normal as every number within this interval rounds to 4
  - Remember that for a normal distribution the probability of a single value is zero so $\mathrm{P}(3.5 \le X_N < 4.5) = \mathrm{P}(3.5 < X_N < 4.5)$

How do I apply continuity corrections?

- Think about what is **largest/smallest integer** that can be included in the inequality for the discrete distribution and then find its **upper/lower bound**
- $P(X = k) \approx P(k - 0.5 < X_N < k + 0.5)$
- $P(X \le k) \approx P(X_N < k + 0.5)$
  - You add 0.5 as you want to include k in the inequality
- $P(X < k) \approx P(X_N < k - 0.5)$
  - You subtract 0.5 as you don't want to include k in the inequality
- $P(X \ge k) \approx P(X_N > k - 0.5)$
  - You subtract 0.5 as you want to include k in the inequality
- $P(X > k) \approx P(X_N > k + 0.5)$
  - You add 0.5 as you don't want to include k in the inequality
- For a closed inequality such as $\mathrm{P}(a < X \le b)$
  - Think about each inequality separately and use above
  - $P(X > a) \approx P(X_N > a + 0.5)$

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- $P(X \leq b) \approx P(X_N < b + 0.5)$
- Combine to give
- $P(a + 0.5 < X_N < b + 0.5)$

How do I approximate a probability?

- **STEP 1**: Find the **mean** and **variance** of the approximating distribution
  - $\mu = np$
  - $\sigma^2 = np(1 - p)$
- **STEP 2**: Apply **continuity corrections** to the inequality
- **STEP 3**: Find the **probability** of the new corrected inequality
  - Find the standard normal probability and use the **table of the normal distribution**
- The probability will not be exact as it is an approximate but provided n is large and p is close to 0.5 then it will be a close approximation
  - To decide if **n is large enough** and if **p is close enough to 0.5** check that:
    - $np > 5$
    - $np > 5$ where $q = 1 - p$

Worked Example

The random variable $X \sim B(1250, 0.4)$.

Use a suitable approximating distribution to approximate $P(485 \leq X \leq 530)$.

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Step 1: Find $X_N \sim N(\mu, \sigma^2)$

$\mu = np = (1250)(0.4) = 500$

$\sigma^2 = np(1-p) = (1250)(0.4)(0.6) = 300$

$\sigma = \sqrt{300}$

Step 2: Apply continuity corrections

$485 \leqslant X \leqslant 530$

Include 485 Include 530

$484.5 \leqslant X_N \leqslant 530.5$

Step 3: Calculate the probability

$P(484.5 \leqslant X_N \leqslant 530.5)$

$= P\left(\frac{484.5 - 500}{\sqrt{300}} \leqslant Z \leqslant \frac{530.5 - 500}{\sqrt{300}}\right)$

$= P(-0.8948... \leqslant Z \leqslant 1.7609...)$

$= \Phi(1.761) - (1 - \Phi(0.895))$

$= 0.9609 - (1 - 0.8147)$

$= 0.7756$

0.775 (3sf)

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Examiner Tips and Tricks

- In the exam, the question will often tell you to use a normal approximation but sometimes you will have to recognise that you should do so for yourself. Look for the conditions mentioned in this revision note, *n* is large, *p* is close to 0.5, *np* > 5 and *nq* > 5.